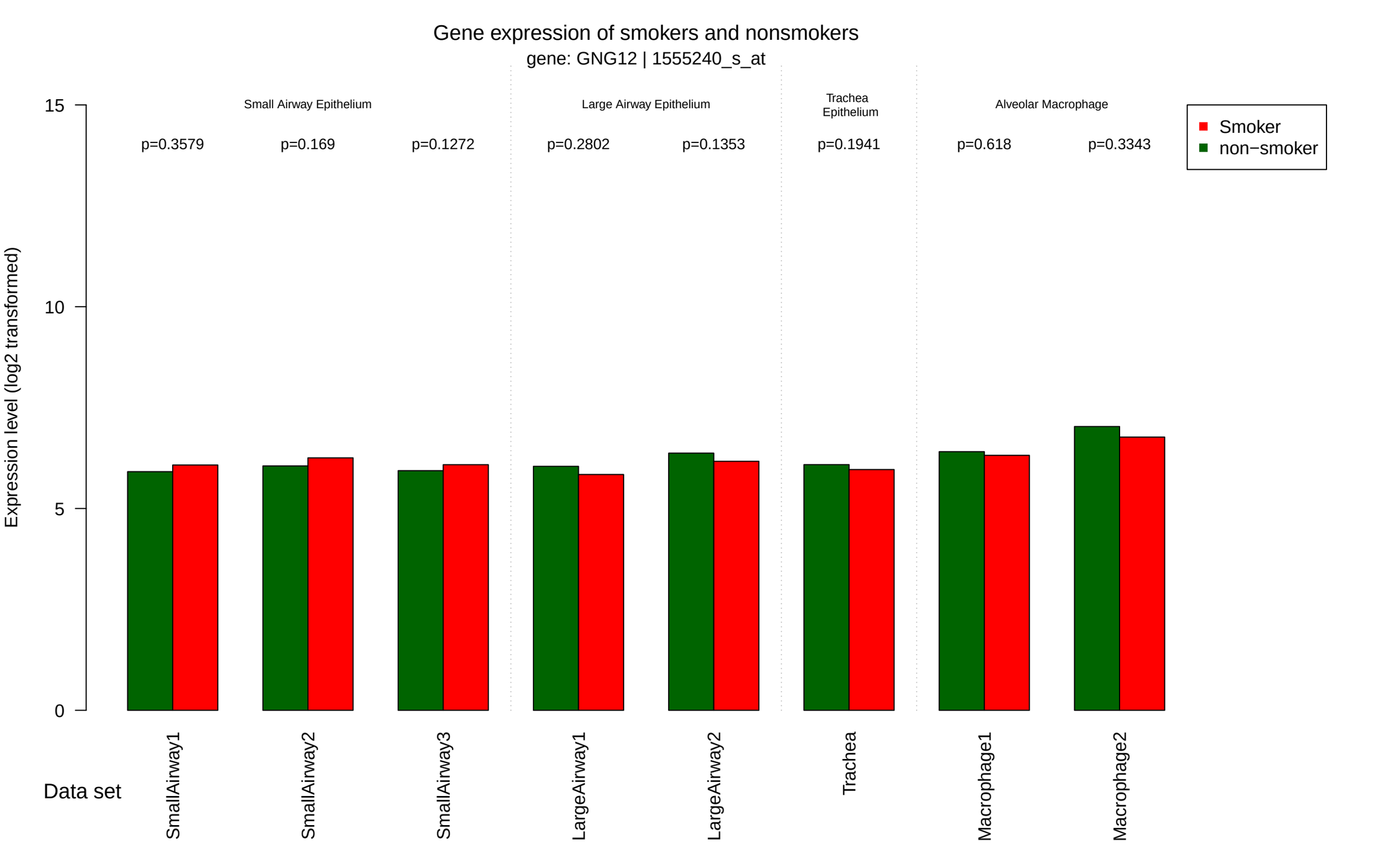
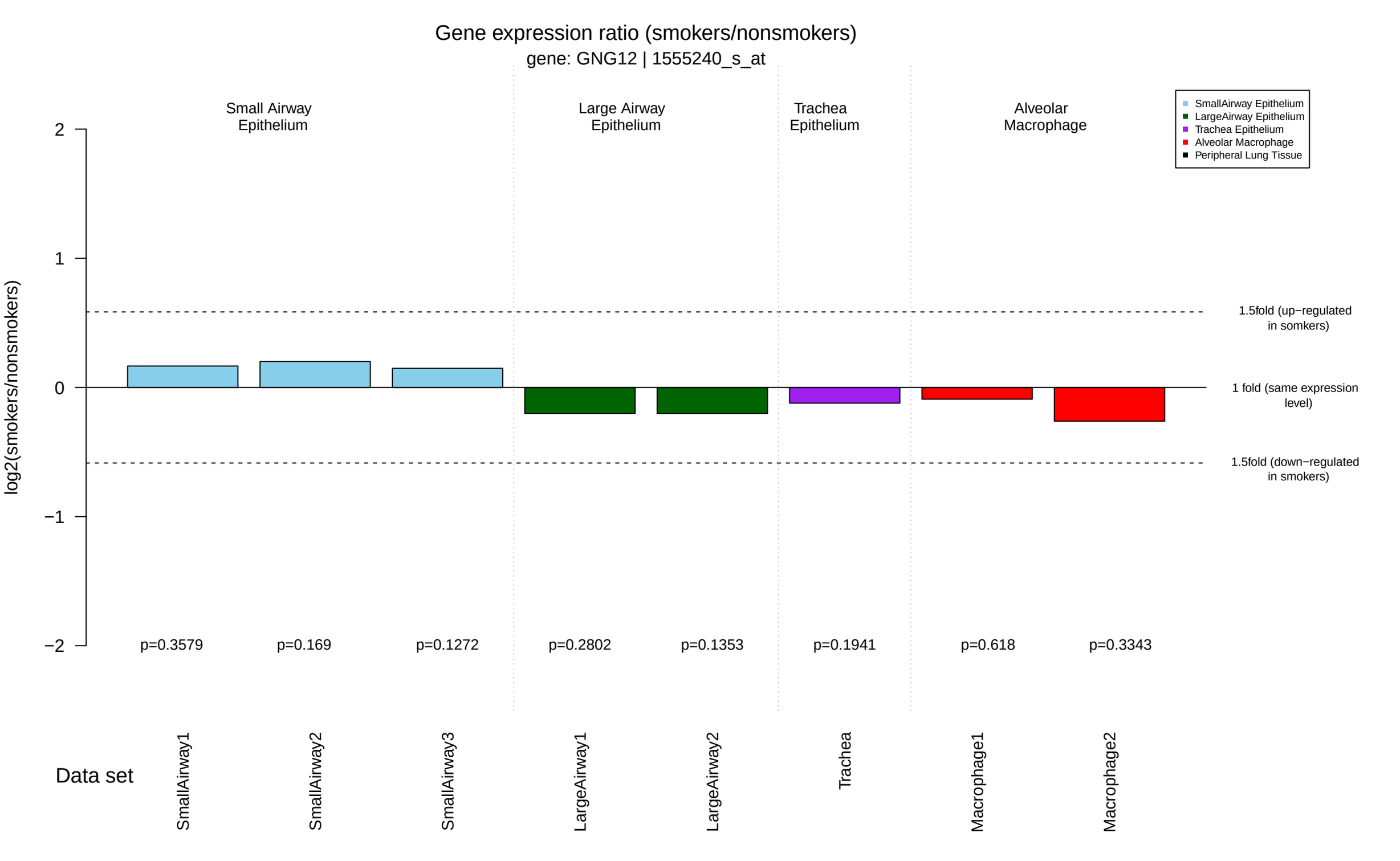
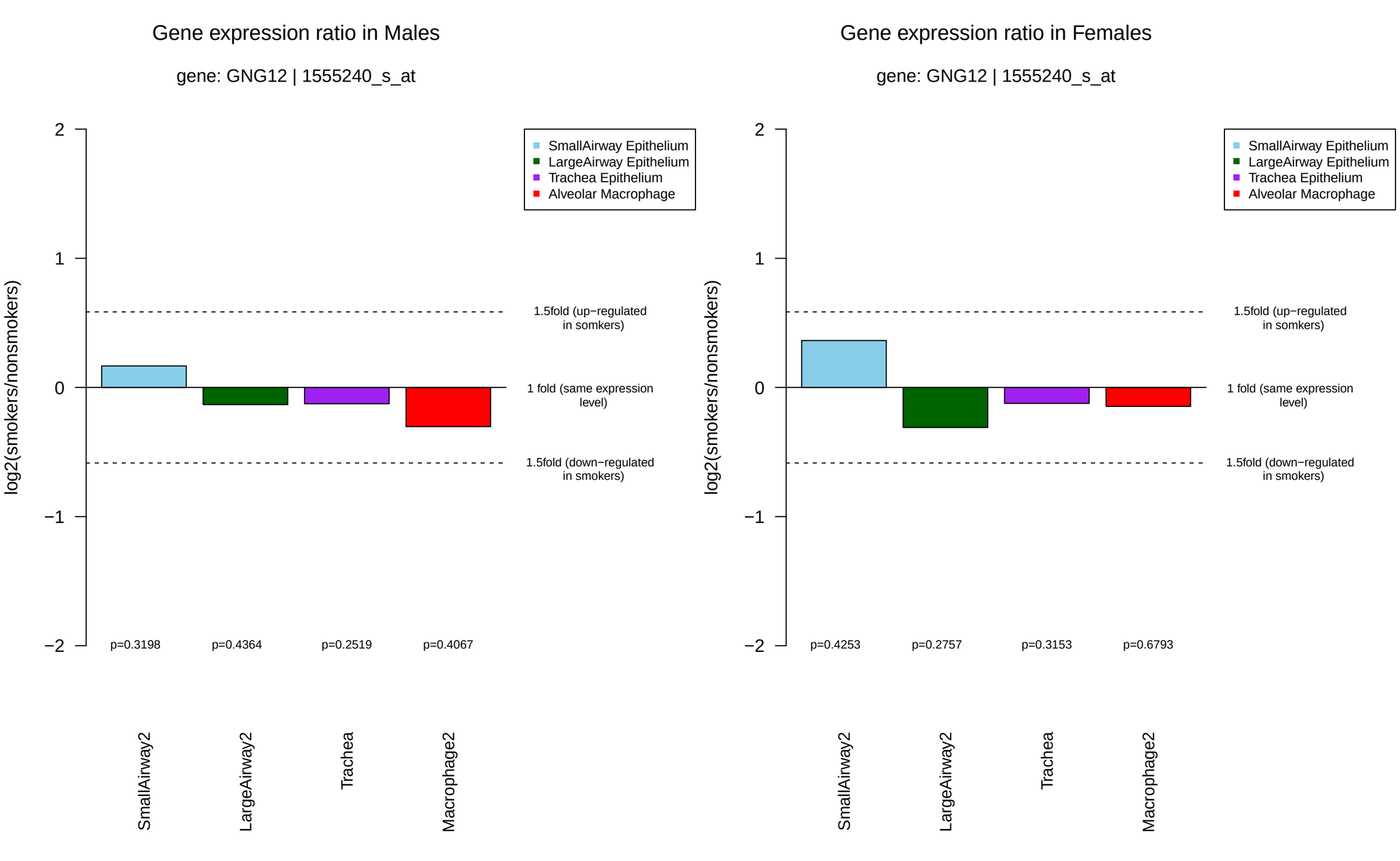
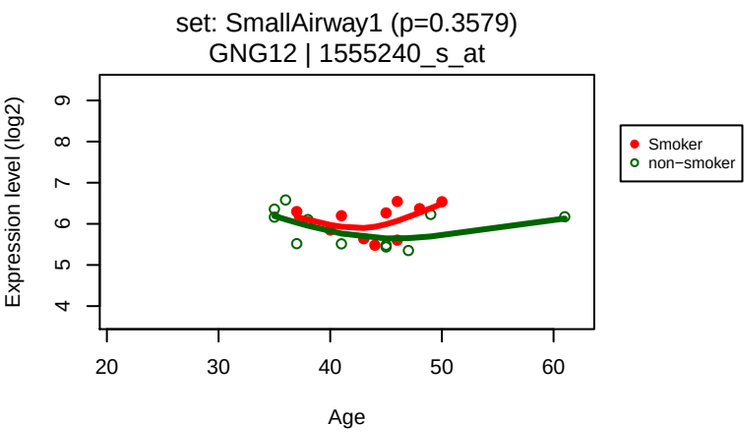
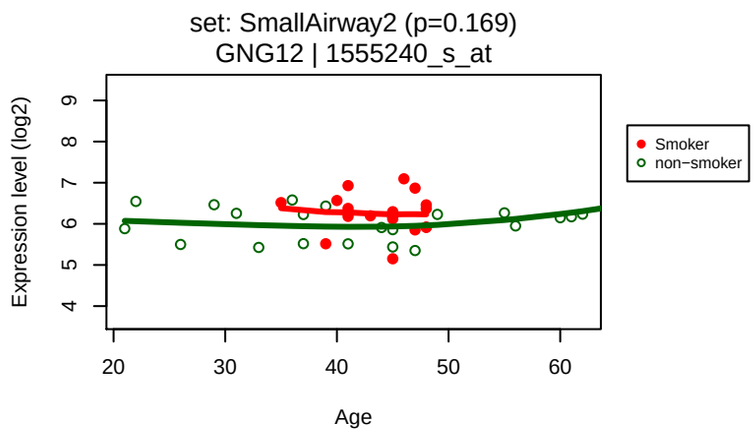
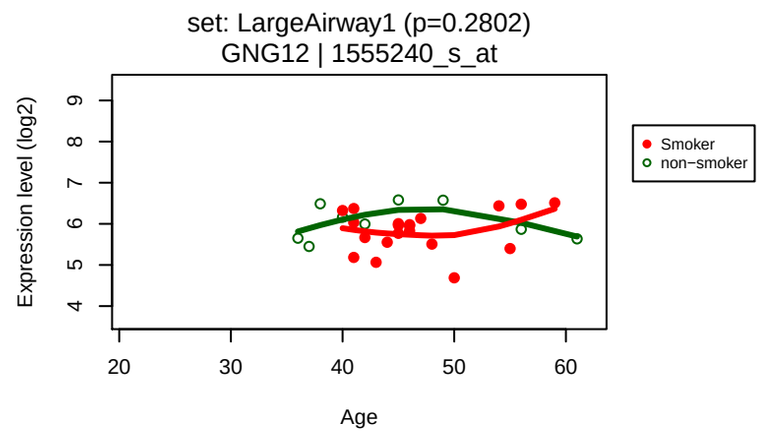

Gene expression of smokers and nonsmokers
gene: GNG12 | 1555240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3579
p=0.169
p=0.1272
p=0.2802
p=0.1353
p=0.1941
p=0.618
p=0.3343
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

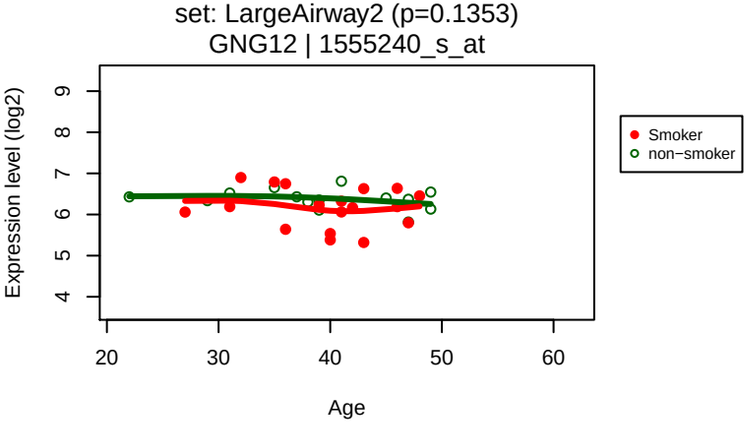

Gene expression ratio (smokers/nonsmokers)
gene: GNG12 | 1555240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3579
p=0.169
p=0.1272
p=0.2802
p=0.1353
p=0.1941
p=0.618
p=0.3343
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

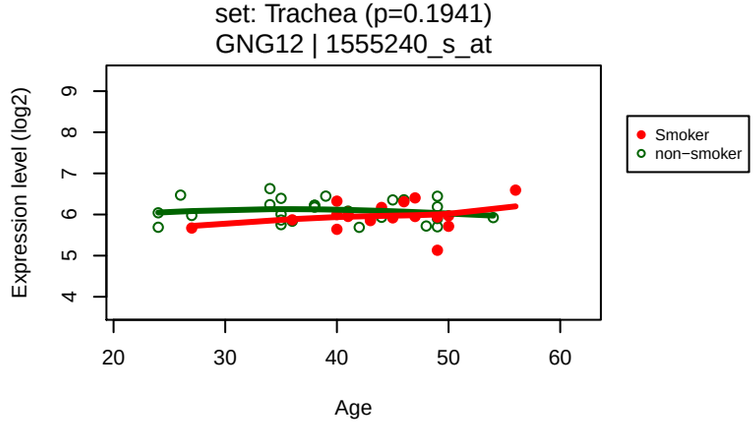

Gene expression ratio in Males
gene: GNG12 | 1555240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3198
p=0.4364
p=0.2519
p=0.4067
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNG12 | 1555240_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4253
p=0.2757
p=0.3153
p=0.6793
SmallAirway2
LargeAirway2
Trachea
Macrophage2

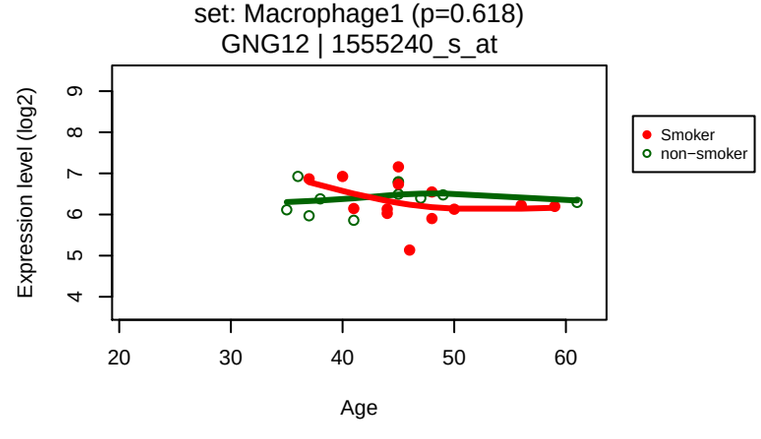

set: SmallAirway1 (p=0.3579)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker

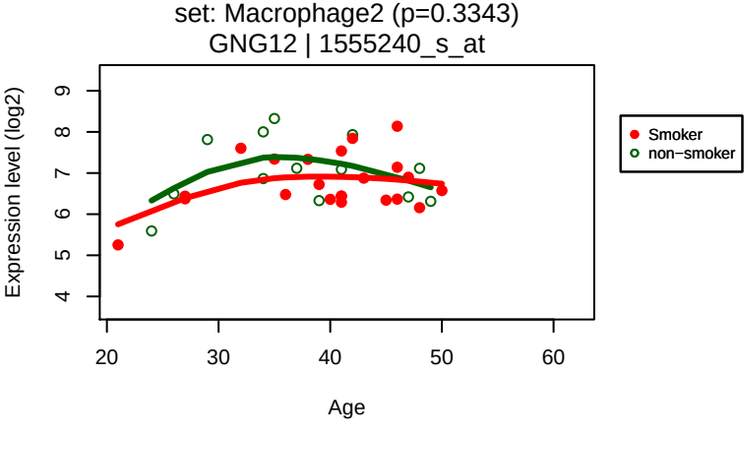

set: SmallAirway2 (p=0.169)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker

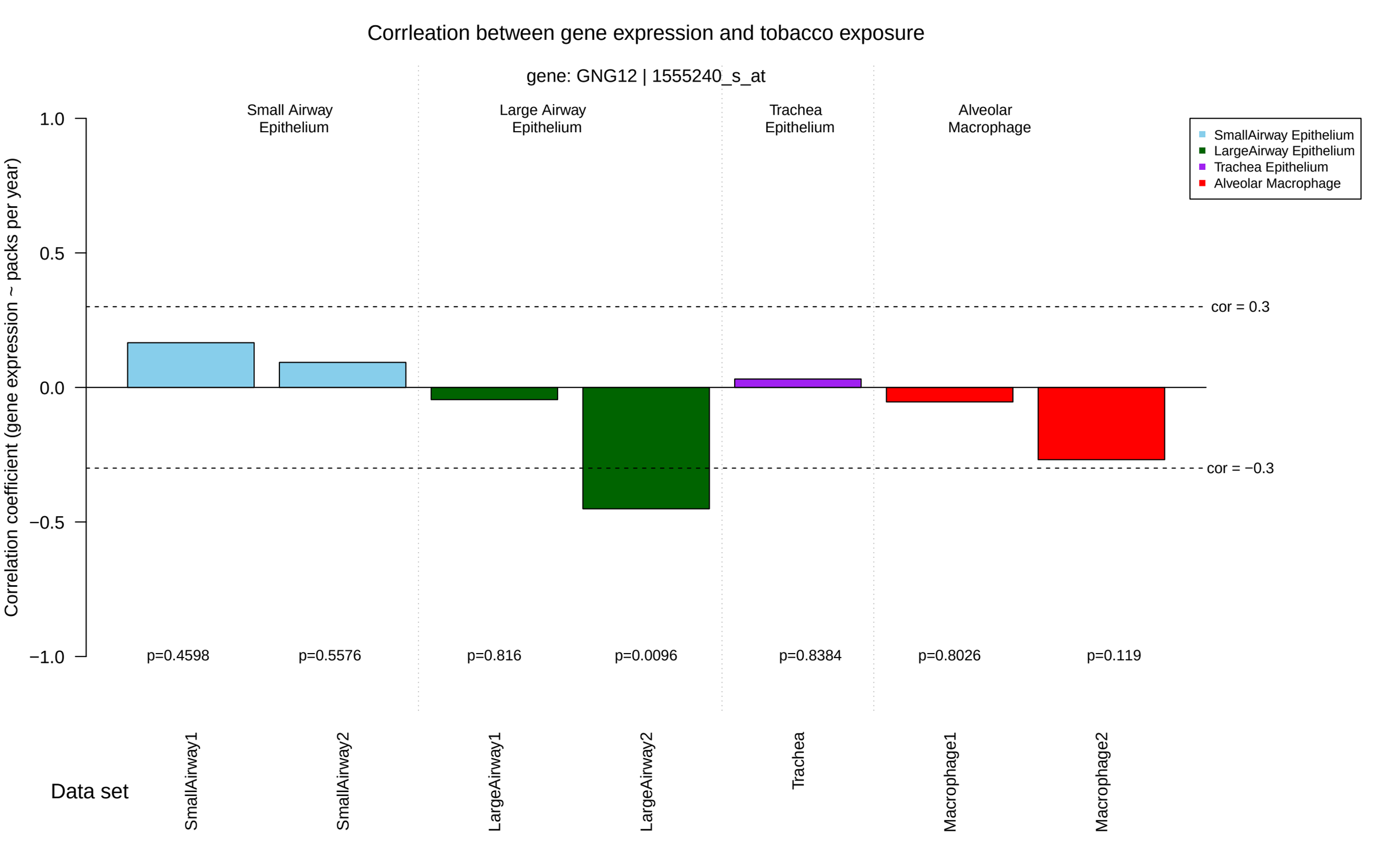

set: LargeAirway1 (p=0.2802)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1353)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1941)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.618)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3343)
GNG12 | 1555240_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNG12 | 1555240_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4598
p=0.5576
p=0.816
p=0.0096
p=0.8384
p=0.8026
p=0.119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2